

SAP Build Beginner's Complete Guide

With Real-Time Practical Examples

What You'll Learn

- What is SAP Build & its three pillars
- SAP Build Apps – No-Code app creation
- SAP Build Process Automation – Workflows & Bots
- SAP Build Work Zone – Digital Workspaces
- Real-time scenario walkthroughs for each tool
- Hands-on exercises & best practices

Beginner · Practical · 2024 Edition

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1. Introduction to SAP Build

SAP Build is SAP's **low-code / no-code platform** that empowers both business users and developers to rapidly create applications, automate processes, and design digital workspaces — all without deep programming knowledge.

The Three Pillars of SAP Build

Pillar	Tool	What It Does	Who Uses It
■ Apps	SAP Build Apps	Create web & mobile applications visually using drag-and-drop	Business Users, Citizen Devs
■ Automation	SAP Build Process Automation	Automate workflows, approvals & repetitive tasks with RPA bots	Process Owners, IT Teams
■ Work Zone	SAP Build Work Zone	Build digital workspaces & employee portals integrating SAP apps	HR, IT, End Users

■ **SAP Build is part of SAP BTP (Business Technology Platform). You can access it at build.cloud.sap — a free trial is available!**

Why SAP Build?

Traditional SAP development requires ABAP or Java expertise and long delivery cycles. SAP Build changes this by letting business users directly participate in building solutions, reducing IT backlog by up to 80% for common use cases.

✓ Speed	Build apps in hours instead of months
✓ No Coding Required	Drag-and-drop UI with pre-built components
✓ SAP Integration	Native connectors to S/4HANA, SuccessFactors, Ariba
✓ Collaboration	Business & IT work together in the same environment
✓ AI-Powered	Joule AI assistant helps build faster with natural language

2. SAP Build Apps – No-Code / Low-Code

SAP Build Apps (formerly AppGyver) is a **visual application builder**. It lets you design pixel-perfect UIs, connect to any REST/OData API, and deploy to web and mobile — all from a browser-based IDE.

Key Components

Component	Description
Canvas	Drag-and-drop workspace where you place UI elements
Component Library	100+ pre-built UI components (buttons, lists, forms, charts)
Data Variables	Bind data from APIs or local storage to your UI automatically
Logic Canvas	Visual flow-based event handling (no code needed)
App Previewer	Instant preview on mobile via SAP Build Apps Preview app
Formula Editor	Excel-like formulas for transformations and conditions

Data Connections in Build Apps

Build Apps connects to data through **Data Resources**. You simply configure the endpoint, and Build Apps auto-generates CRUD operations:

- **OData V2/V4** — Direct connection to SAP S/4HANA, BTP services
- **REST API** — Any external service with a REST API
- **SAP BTP Destination** — Secure managed connections stored in BTP
- **On-Device Storage** — Local app data (offline capability)
- **SAP Integration Suite** — Enterprise integrations via iFlows

■ **Real-World Use Cases:** Employee self-service portals, expense reporting, asset inspection apps, site visit checklists, customer visit reports.

3. Real-Time Example: Leave Request App

■ Scenario: HR wants employees to submit leave requests via a mobile app that connects to SAP SuccessFactors — without writing a single line of code.

Step-by-Step Walkthrough

1	Open SAP Build Apps Go to build.cloud.sap → Click 'Create' → Choose 'Build an Application'
2	Name Your Project Enter 'Leave Request App' → Select 'Web & Mobile Application'
3	Set Up Data Resource Go to Data tab → Add REST API Resource → Enter SuccessFactors OData URL for Time Off
4	Design the Home Screen Drag a 'Form' component onto canvas → Add fields: Employee ID, Start Date, End Date, Leave Type, Reason
5	Bind Data Select each field → In Properties panel, bind to the corresponding data variable from SuccessFactors
6	Add Submit Logic Click Submit button → Open Logic Canvas → Add 'Create Record' node → Connect to SuccessFactors → Add 'Toast' for success message
7	Add Validation Use Formula Editor: IF(IS_EMPTY(leaveType), 'Please select leave type', '')
8	Preview & Test Click Play button → Scan QR code with SAP Build Preview app on your phone
9	Deploy Click Publish → Choose 'Web App' or 'Mobile' → Share URL with employees

What the App Looks Like

Home Screen	Confirmation Screen
[Leave Request Form] Employee: John Doe Start Date: [Date Picker] End Date: [Date Picker] Type: [Annual ▼] Reason: [Text Area] [Submit Request]	[✓ Request Submitted] Leave ID: LV-20240315 Status: Pending Approval Duration: 3 Days Approver: Jane Smith [View My Requests] [Submit Another]

Fig 1: Leave Request App – Home & Confirmation Screens

■ Time to build this app: ~2 hours for a beginner. The same app in traditional ABAP development would take 2–3 weeks.

4. SAP Build Process Automation

SAP Build Process Automation (formerly SAP Workflow Management + Intelligent RPA) lets you automate business processes using **workflows, forms, decisions, and bots** — all visually designed.

Core Capabilities

Capability	Description	Example
Process Builder	Drag-and-drop workflow designer	Multi-step approval chain
Forms	Auto-generated approval/task forms	Manager approve/reject UI
Decisions	Business rules engine	Route by amount: >\$10K needs CFO
RPA Bots	Automate desktop/web tasks	Copy data from legacy system
Events	Trigger automation from SAP events	New PO created → start approval
Monitoring	Real-time process tracking dashboard	See stuck instances

Process Automation Architecture

- **Trigger:** Event from S/4HANA, form submit, API call, schedule
- **Process Flow:** Sequence of tasks, approvals, decisions, parallel branches
- **Human Tasks:** Forms sent to users via email or Work Zone inbox
- **Automated Tasks:** API calls, RPA bots, decision rules run automatically
- **Outcome:** Update record, send notification, start another process

5. Real-Time Example: Invoice Approval Workflow

■ **Scenario:** Finance team needs a 3-level invoice approval workflow. Invoices under \$5K need 1 approver; \$5K–\$50K need manager + finance; above \$50K additionally require CFO sign-off.

Workflow Design Steps

1	Open Process Builder SAP Build → Create → Build an Automated Process → Name it 'Invoice Approval'
2	Define Trigger Add Trigger → 'API Trigger' (called when invoice created in S/4HANA)
3	Add Approval Step 1 Drag 'Approval' task → Configure form with: Invoice #, Vendor, Amount, Line Items → Assign to Line Manager
4	Add Decision Rule Drag 'Decision' → Create rule table: Amount < 5000 → End, 5000-50000 → Finance Review, > 50000 → CFO Review
5	Add Approval Step 2 Add Approval task → Finance Manager → Inputs: Previous decision + Invoice details
6	Add Conditional CFO Step Add Gateway → If Amount > 50000 → Route to CFO Approval step
7	Post-Approval Action Add 'Call API' action → Update invoice status in S/4HANA via REST API
8	Set Up Notifications Add Mail steps → Notify requester at each stage with approval status
9	Activate & Monitor Deploy process → Test with sample invoice → Monitor in Process Monitoring dashboard

Process Flow Diagram

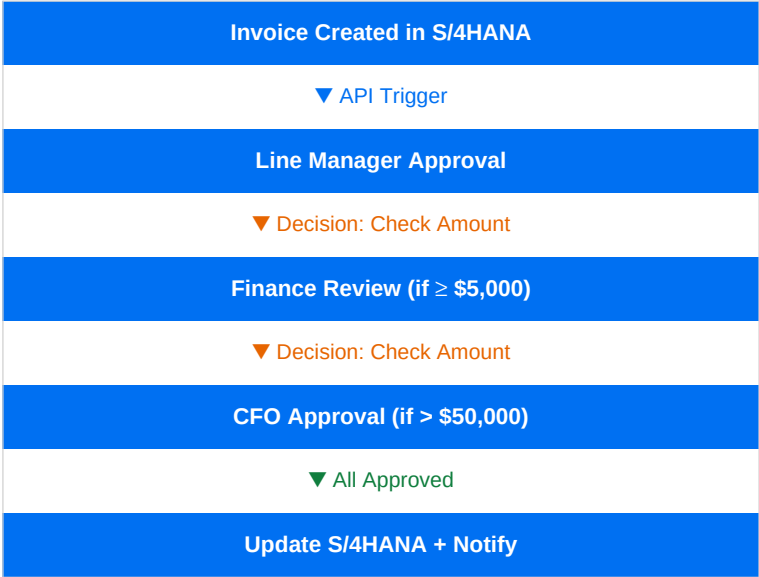


Fig 2: Invoice Approval Workflow – Process Flow

6. SAP Build Work Zone

SAP Build Work Zone is a **digital workspace platform** that brings together SAP applications, third-party tools, and custom apps into a single unified experience — like a personalized intranet powered by SAP.

Key Features

Feature	Description
Workpages	Customizable dashboard pages with widgets, tiles & feeds
Business Apps Integration	Embed SAP Fiori apps, Build Apps, external URLs
Notifications	Unified inbox for workflow tasks and alerts
Workspace Templates	Pre-built templates for HR, Finance, IT portals
Role-Based Access	Different content shown to different user roles
Mobile Support	Responsive design + native iOS/Android app
Content Management	News, announcements, documents, KB articles

Two Editions

- ◆ **Standard Edition:** Basic portal with SAP Fiori launchpad replacement. Ideal for SAP app aggregation.
- ◆ **Advanced Edition:** Full digital workplace with HR content, communities, and advanced customization.

7. Real-Time Example: HR Digital Workspace

■ **Scenario:** HR team wants a single portal where employees can: view pay slips, apply for leave, see company news, access training — all in one place, integrated with SuccessFactors.

Building the HR Portal

1	Access Work Zone BTP Cockpit → Instances → SAP Build Work Zone → Open Admin Console
2	Create a Site Site Manager → Create Site → Name: 'HR Employee Portal' → Choose 'HR Template'
3	Add a Workpage Pages → New Page → 'Employee Self-Service' → Set as homepage for 'Employee' role
4	Add Content Tiles Drag tiles: 'My Pay Slips' (links to SuccessFactors), 'Apply for Leave' (links to Leave App), 'My Team' (SuccessFactors Org Chart)
5	Add News Widget Drag 'News Feed' widget → Connect to Work Zone Content channel 'HR Announcements'
6	Add Pending Tasks Drag 'My Inbox' widget → Shows pending approvals from Process Automation
7	Configure Navigation Set up left navigation bar: Home, My Profile, Benefits, Learning, IT Help
8	Assign Roles Role-Based Navigation: Managers see additional 'Team Management' section
9	Test & Publish Preview as Employee role → Check all tiles load → Publish site → Share URL

Portal Layout Preview

HR Employee Portal – Home Dashboard		
■ My Tasks (3 pending) ■■■■■■■■■■■■■■■■■■■■ ■ Approve Leave: John ■ Review Goal: Q3 ■ Sign Policy Doc	■ HR News ■■■■■■■■■■■■■■■■■■■■ ■ New WFH Policy ■ Benefits Enrolment ■ Townhall: Dec 15	■ Quick Actions ■■■■■■■■■■■■■■■■■■■■ ■ Apply Leave ■ View Payslip ■ Update Profile ■ Submit Expense

Fig 3: HR Employee Portal built with SAP Build Work Zone

8. Comparison & When to Use What

Use Case	Recommended Tool	Why
Build a mobile app for field workers	SAP Build Apps	Visual UI builder, offline support
Automate invoice approval with 3 levels	Process Automation	Workflow + decision + forms
Create employee onboarding portal	Work Zone	Aggregate apps + content + tasks
Replace Excel-based data entry	SAP Build Apps	Forms + OData integration
Automate data migration between systems	Process Automation (RPA)	Bot can navigate legacy UIs
Single launchpad for all SAP apps	Work Zone Standard	Fiori launchpad replacement
Expense report with manager approval	Apps + Process Automation	App for input, workflow for approval
Chatbot for IT helpdesk	SAP Build Apps + AI	Conversational UI with AI backend

■ Often you combine all three! Example: Build Apps (UI) → triggers Process Automation (workflow) → surfaces tasks in Work Zone (portal). This is called the SAP Build 'Power Combo'.

9. Best Practices & Tips for Beginners

■ Getting Started

- Start with SAP Build's free trial on BTP Trial account
- Complete the SAP Learning Journey: 'Become an SAP Build Apps Developer'
- Join the SAP Build Community at community.sap.com for templates & help
- Use the pre-built templates instead of starting from scratch

■ Build Apps Tips

- Always set up Data Resources before designing your UI
- Use App Variables for global state; Page Variables for local state
- Test on mobile early — the preview app is free on iOS and Android
- Use the Formula Editor's autocomplete — it shows all available variables

■ Process Automation Tips

- Map out your process on paper before building in the tool
- Always add error handling to API call tasks
- Use the 'Process Variants' feature to handle exceptions without duplicating
- Monitor the 'Acquired' status — instances stuck here need attention

■ Work Zone Tips

- Use Role-Based Access from the start — retrofitting is difficult
- Test the mobile experience before going live
- Use SAP's pre-built content packages for HR, Finance to save time
- Set a dedicated 'site admin' user — don't use your personal account

10. Quick-Reference Glossary

BTP	SAP Business Technology Platform – cloud platform hosting SAP Build
Citizen Developer	Business user who builds apps without coding skills
OData	Open Data Protocol – SAP's primary API standard for S/4HANA
LCNC	Low-Code / No-Code – development approach requiring minimal coding
RPA Bot	Software robot that automates repetitive desktop/web tasks
Workflow Instance	One running execution of a process (e.g., one invoice approval)
Fiori	SAP's UX design system – Work Zone hosts Fiori apps
Destination	BTP-managed secure connection to a backend system
Workpage	A configurable page in Work Zone with widgets and content
Logic Canvas	Visual event-flow editor in Build Apps (no code)
Decision Table	Business rules configured as spreadsheet-like tables
Site	A complete Work Zone portal with pages, navigation, and branding
Subaccount	BTP organizational unit where SAP Build services are enabled
S/4HANA	SAP's flagship ERP system that SAP Build integrates with
iFlow	Integration flow in SAP Integration Suite connecting systems

■ Ready to Start?

1. Sign up for BTP Trial: account.hanatrial.ondemand.com
2. Enable SAP Build Apps service in your subaccount
3. Take the free course: SAP Learning Center → search 'SAP Build'
4. Join the community: community.sap.com → tag 'SAP Build'
5. Explore templates: build.cloud.sap → Template Gallery